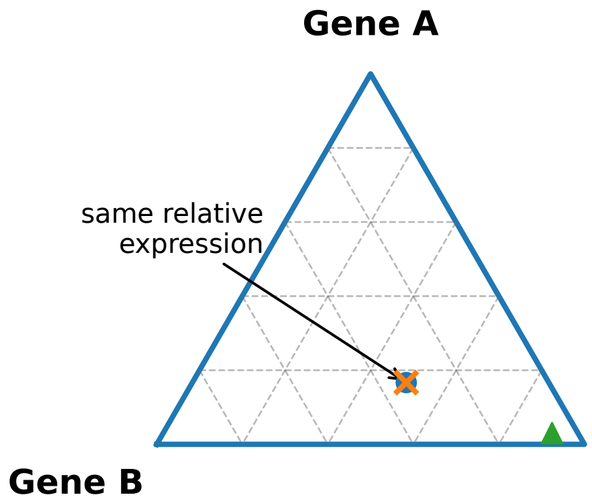
clrcycle workflow

1. Gene expression in CLR space



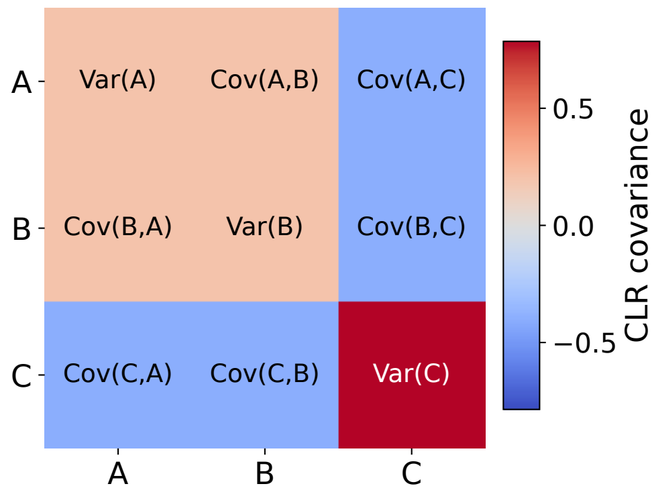
Cell 1: (10, 20, 30)
Cell 2: (100, 200, 300)
Cell 3: (10, 20, 300)

$$z_{ig} = \log \frac{x_{ig}}{g(\mathbf{x}_i)}$$

$$g(\mathbf{x}_i) = \left(\prod_g x_{ig} \right)^{1/G}$$

2. Gene covariance

Σ_{genes} **in CLR space**



3. Maximize first Fourier variance

$$F(\pi) = \mathbf{c}^T \Sigma_\pi \mathbf{c} + \mathbf{s}^T \Sigma_\pi \mathbf{s}$$

π: candidate gene order

Σ_π: CLR covariance reordered by π

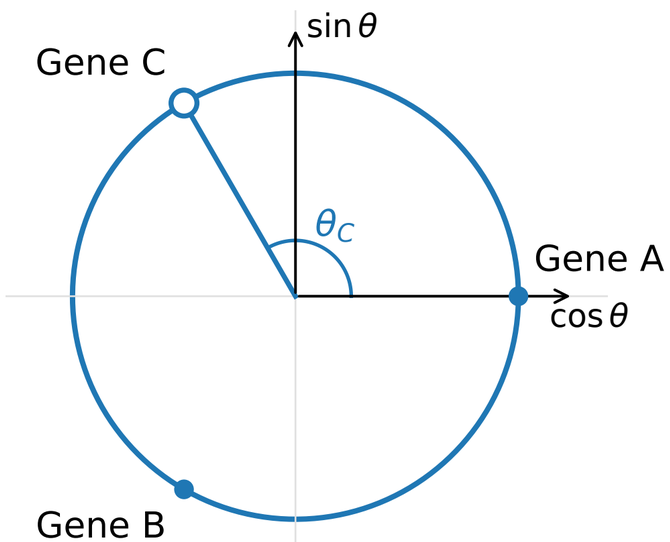
$$\theta_g = 2\pi(g-1)/G$$

$$c_g = \sqrt{\frac{2}{G}} \cos\theta_g, \qquad s_g = \sqrt{\frac{2}{G}} \sin\theta_g$$

$$\hat{\pi} = \arg\max_\pi F(\pi)$$

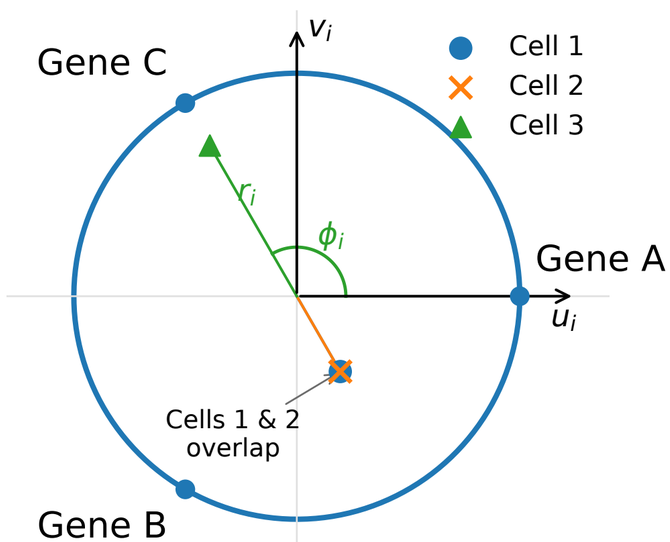
$\hat{\pi}$ is the optimal gene order

4. Gene ordered circular coordinates



$$\theta_C = \frac{2\pi(2-1)}{3} = \frac{2\pi}{3}$$

5. Project cells onto joint feature-sample geometry



$$u_i = \sum_g Z_{i,\hat{\pi}(g)} c_g \qquad v_i = \sum_g Z_{i,\hat{\pi}(g)} s_g$$